Practice Paper 2: Long Answer Questions

Q1. (Matrices) Solve the system of equations using matrix inversion: $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 4y + 9z = 36$.

Q2. (Calculus) Evaluate the integral: $\int x * \sin(x)\, dx$ using Integration by Parts.

Q3. (AP/GP) The sum of three numbers in a GP is 35, and their product is 1000. Find the numbers.

Q4. (Probability) Two dice are thrown. Find the probability distribution of the sum of numbers appearing on them.

Q5. (Coordinate Geo) Find the equation of the parabola with focus (0, 2) and directrix $y = -2$.

Q6. Let $g(x)$ be the function defined by:

$$g(x)=\begin{cases}\frac{\sin (2x)}{x}+k,&x<0\\ 3x^{2}+m,&0\le x\le 2\\ \frac{x^{2}-4}{x-2},&x>2\end{cases}$$

where k and m are constants.

Part A: Find the Value of m

Find the value of m such that $g(x)$ is continuous at $x = 2$. Justify your answer using the definition of continuity.